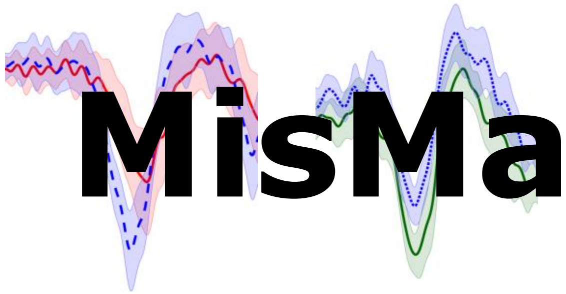
Asymmetries in the MisMatch Negativity: not just phonology



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How the MMN is elicited

a a a a i a a a i a a a a a a i …

a a a a i a a a i a a a a a a i …

deviants

a a a a i a a a i a a a a a a i …

deviants

standards

i i i i a i i i a i i i i i i a …

a a a a i a a a i a a a a a a i …

deviants

standards

i i i i a i i i a i i i i i i a …

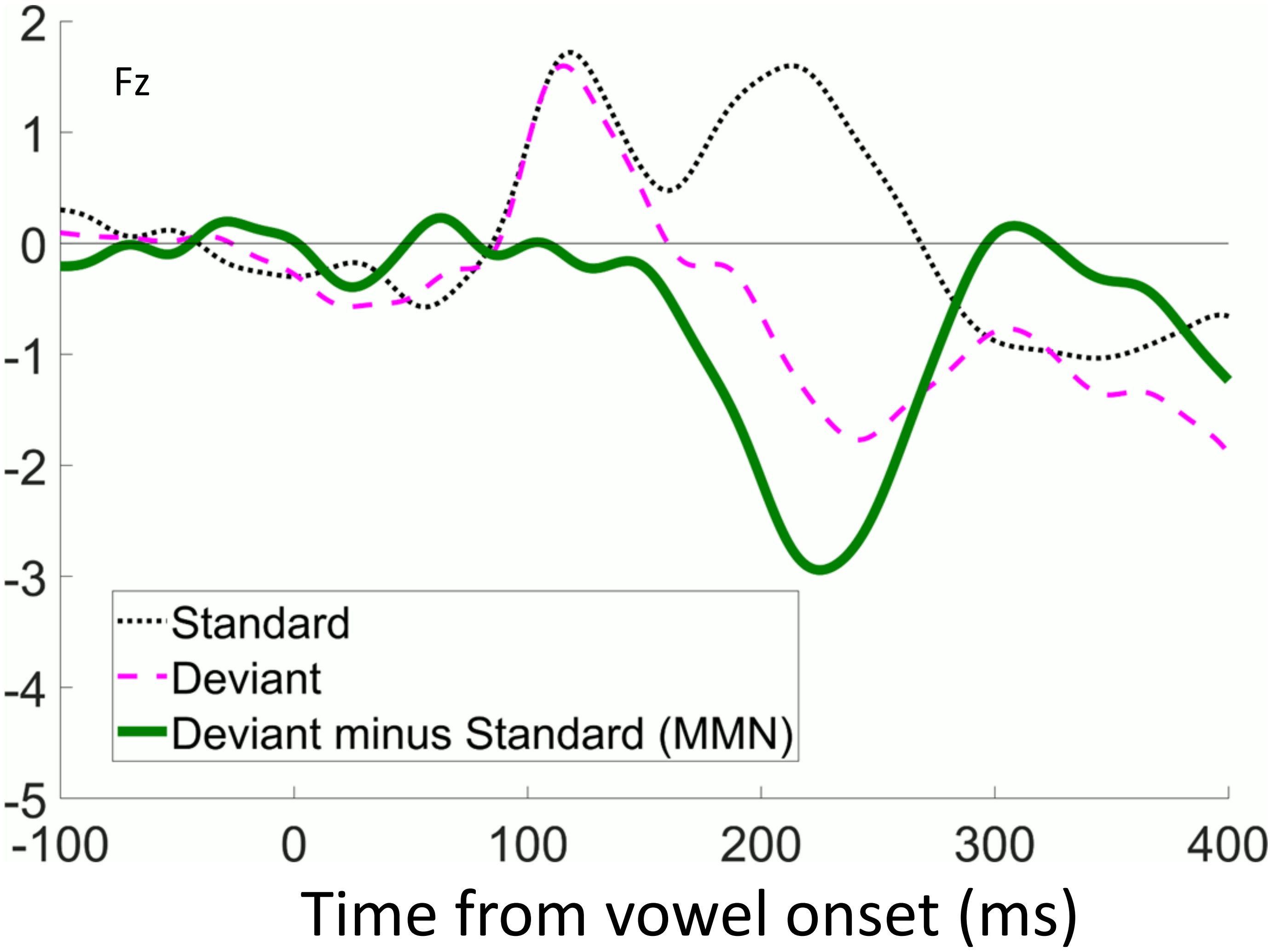


Figure adapted from Schluter, Politzer-Ahles, & Almeida (2016)

a a a a i a a a i a a a a a a i …

deviants

standards

i i i i a i i i a i i i i i i a …

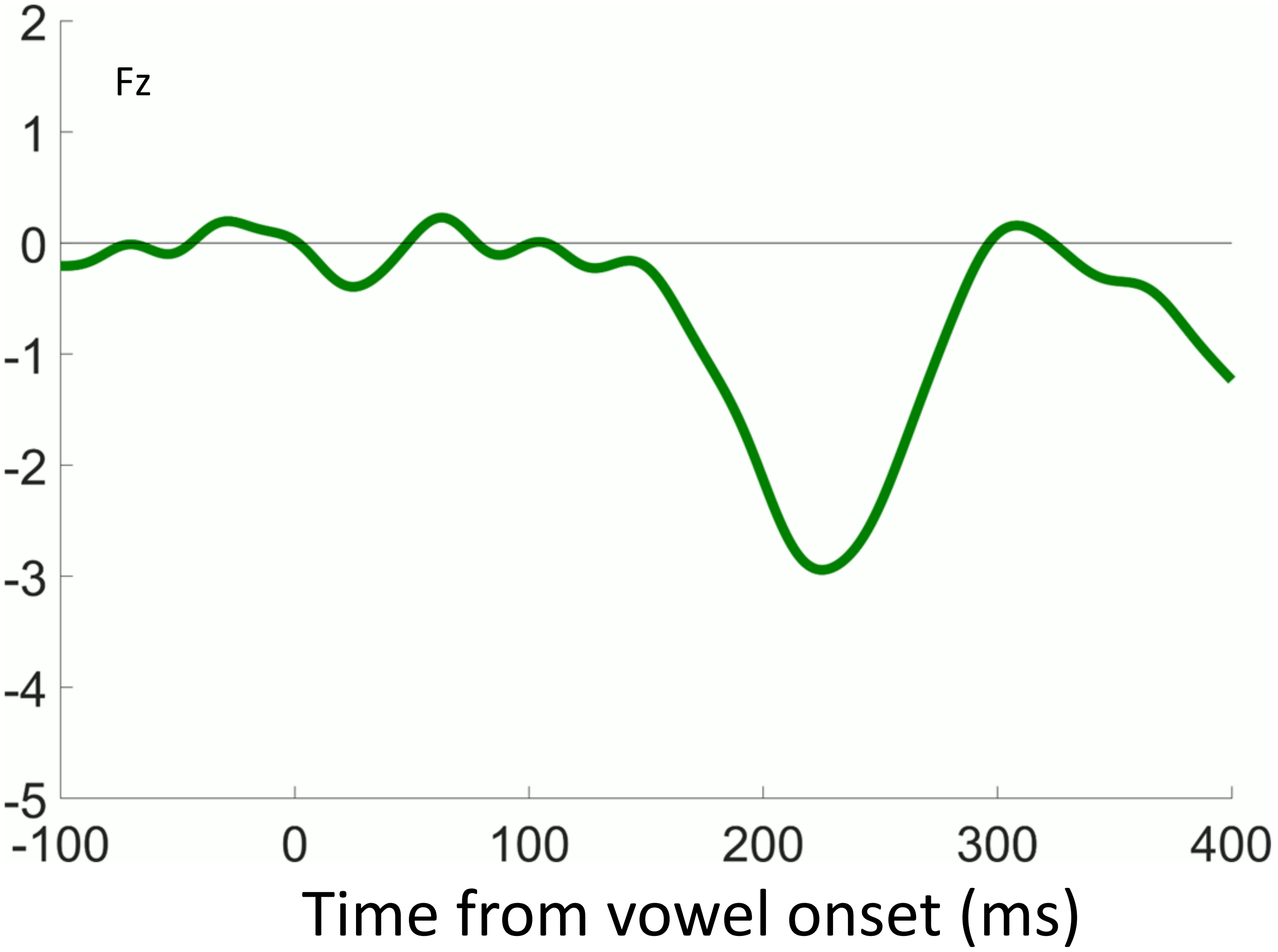


Figure adapted from Schluter, Politzer-Ahles, & Almeida (2016)

The mismatch negativity

a a a a i a a a a i a a a a a a a i …

deviants

standards

i i i i a i i i i a i i i i i i i a …

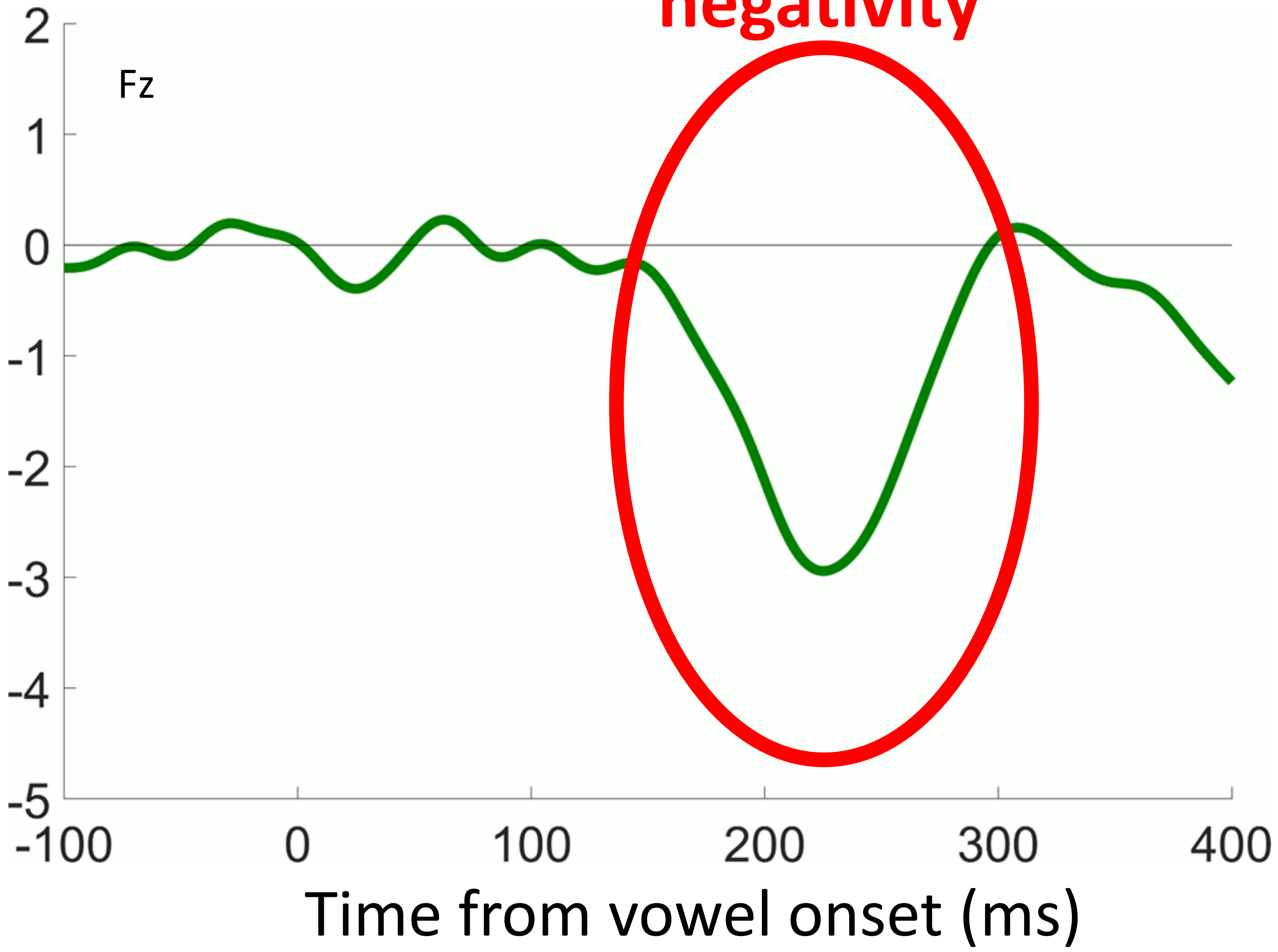


Figure adapted from Schluter, Politzer-Ahles, & Almeida (2016)

MMN requires a "standard" category

- *a a a a a a **i** a a a a a **i** a a a **i**…*

  → will elicit MMN

- *a **i** a a **i** a **i** a **i** **i** a a **i** a **i** a **i** …*

  → will **NOT** elicit MMN

- *u a ɛ ɪ ɔ ʌ **i** a ʌ ʊ æ ɛ **i** ɔ ʊ u **i**…*

  → will **NOT** elicit MMN

ard” category

ard” category

MMN for *abstract* category contrasts

a a a a i a a a i a a a a a a i …

MMN with no reliable physical cue

gave chose bled swore get clung pled grew met pave brought drew sang thank

MMN with no reliable physical cue

gave chose bled swore ***get*** *clung*
pled grew met ***pave*** *brought*
drew sang ***thank***

Politzer-Ahles & Jap (2024)

MMN with no reliable physical cue

*gave chose bled swore **get** clung*
*pled grew met **pave** brought*
*drew sang **thank***

- Present-tense deviants:
  pave, ***get***, ***thank***

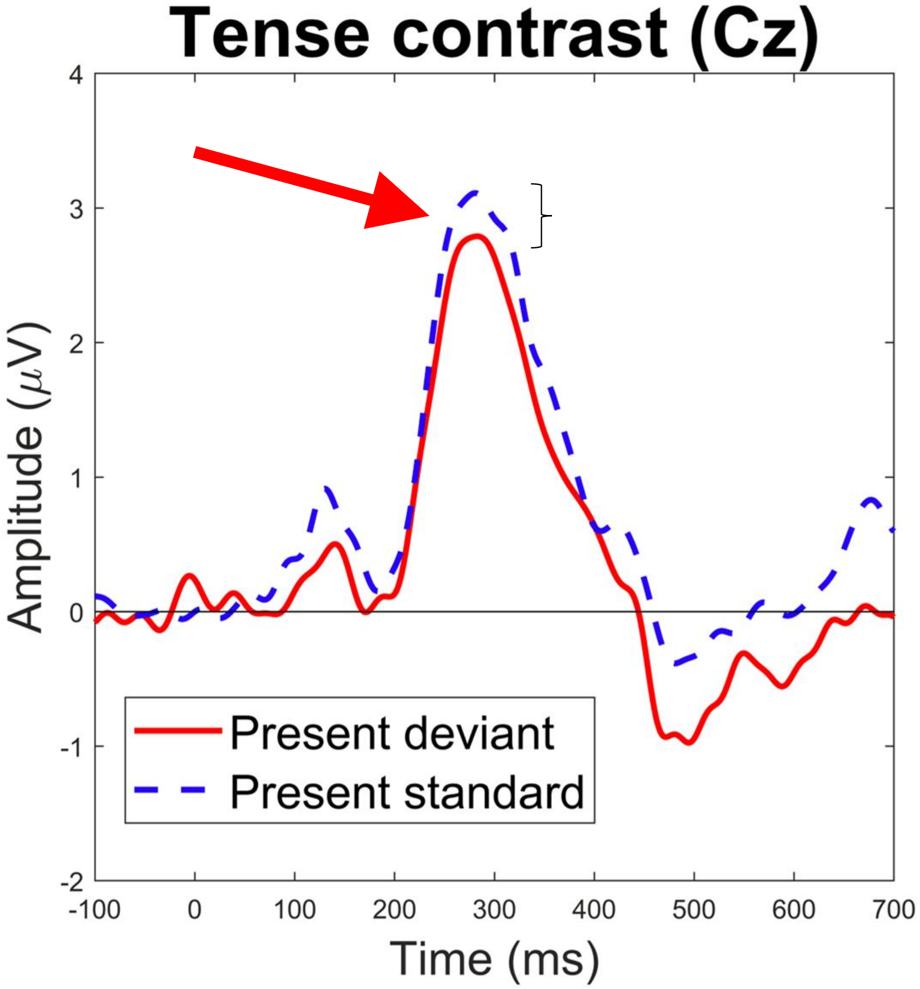
- Past-tense standards:
  gave, *met*, *sank*, etc.

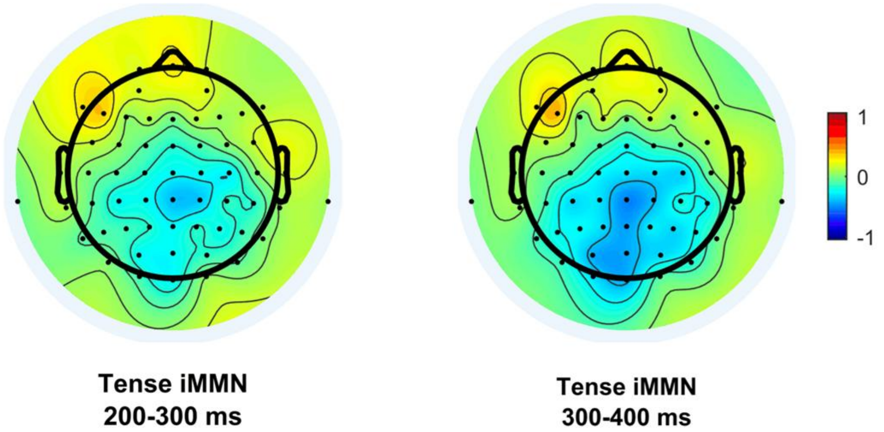
Politzer-Ahles & Jap (2024)

MMN with no reliable physical cue

gave chose bled swore ***get*** *clung pled grew met* ***pave*** *brought drew sang* ***thank***

- Present-tense deviants: ***pave***, ***get***, ***thank***
- Past-tense standards: *gave*, *met*, *sank*, etc.





Politzer-Ahles & Jap (2024)

Interim summary

- The MMN isn't just a detector of low-level physical contrasts; it also detects contrasts between abstract linguistic categories

MMN asymmetries

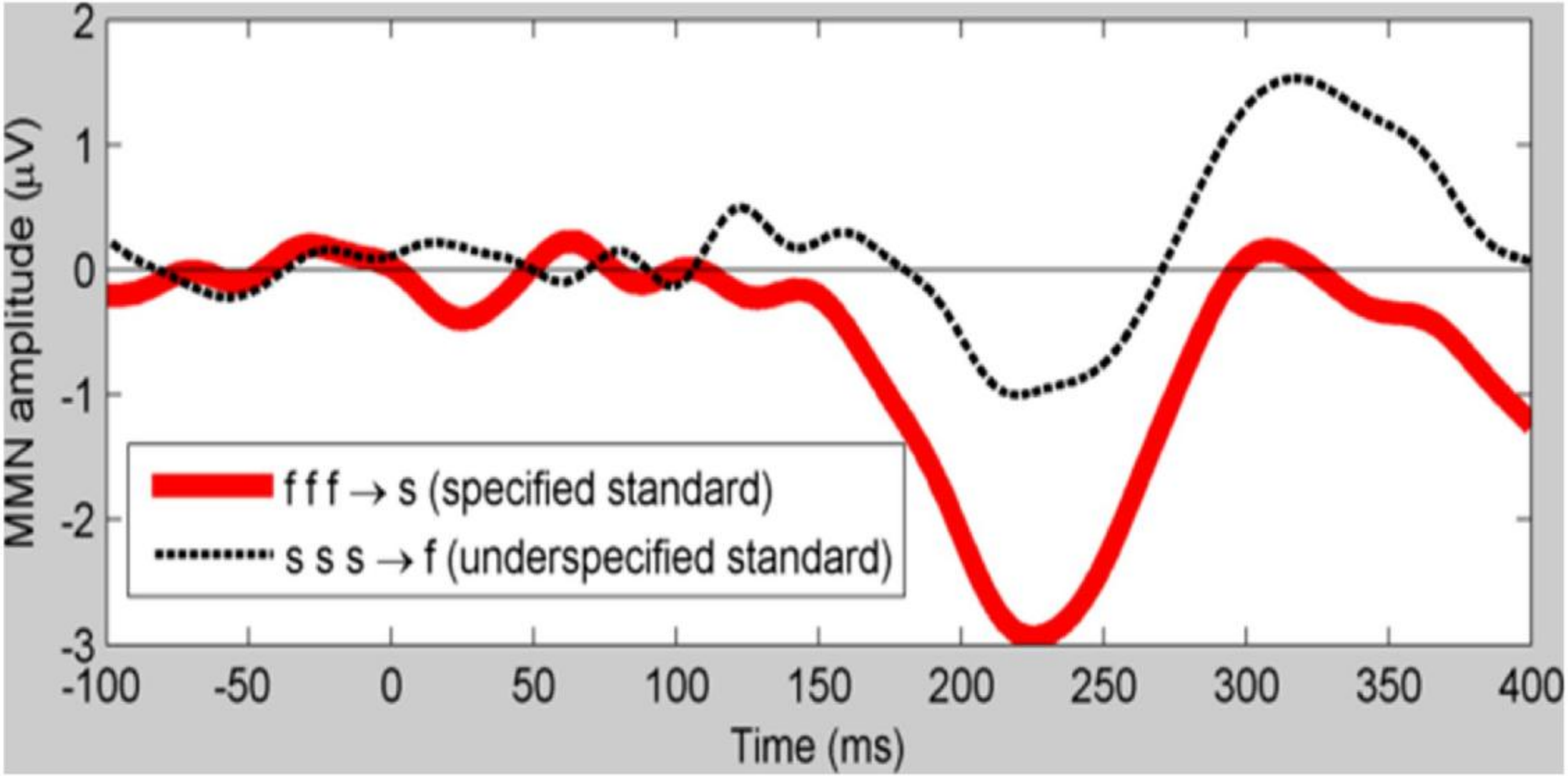


Figure adapted from Schluter, Politzer-Ahles, & Almeida (2016)

The present study

The present study

- We know we get MMN asymmetries for *phonological* contrasts where one category is underspecified

- Does that also happen for *non-phonological* contrasts?

- Test case: the present-past tense contrasts from Politzer-Ahles & Jap (2024) (N=60)

Contrasts

- Present-tense deviant, past-tense standards
  gave chose bled swore ***get*** *clung pled grew met* ***pave*** *brought drew sang* ***thank*** …

- Past-tense deviants, present-tense standards
  pave choose bleed swear ***met*** *cling plead grow get* ***gave*** *bring draw sing* ***sank*** …

Contrasts

Fully specified, marked

- Present-tense deviant, past-tense standards
  gave chose bled swore ***get*** *clung pled grew met* ***pave*** *brought drew sang* ***thank*** …

- Past-tense deviants, present-tense standards
  pave choose bleed swear ***met*** *cling plead grow get* ***gave*** *bring draw sing* ***sank*** …

Defaulty, underspecified?

Contrasts

Fully specified, marked

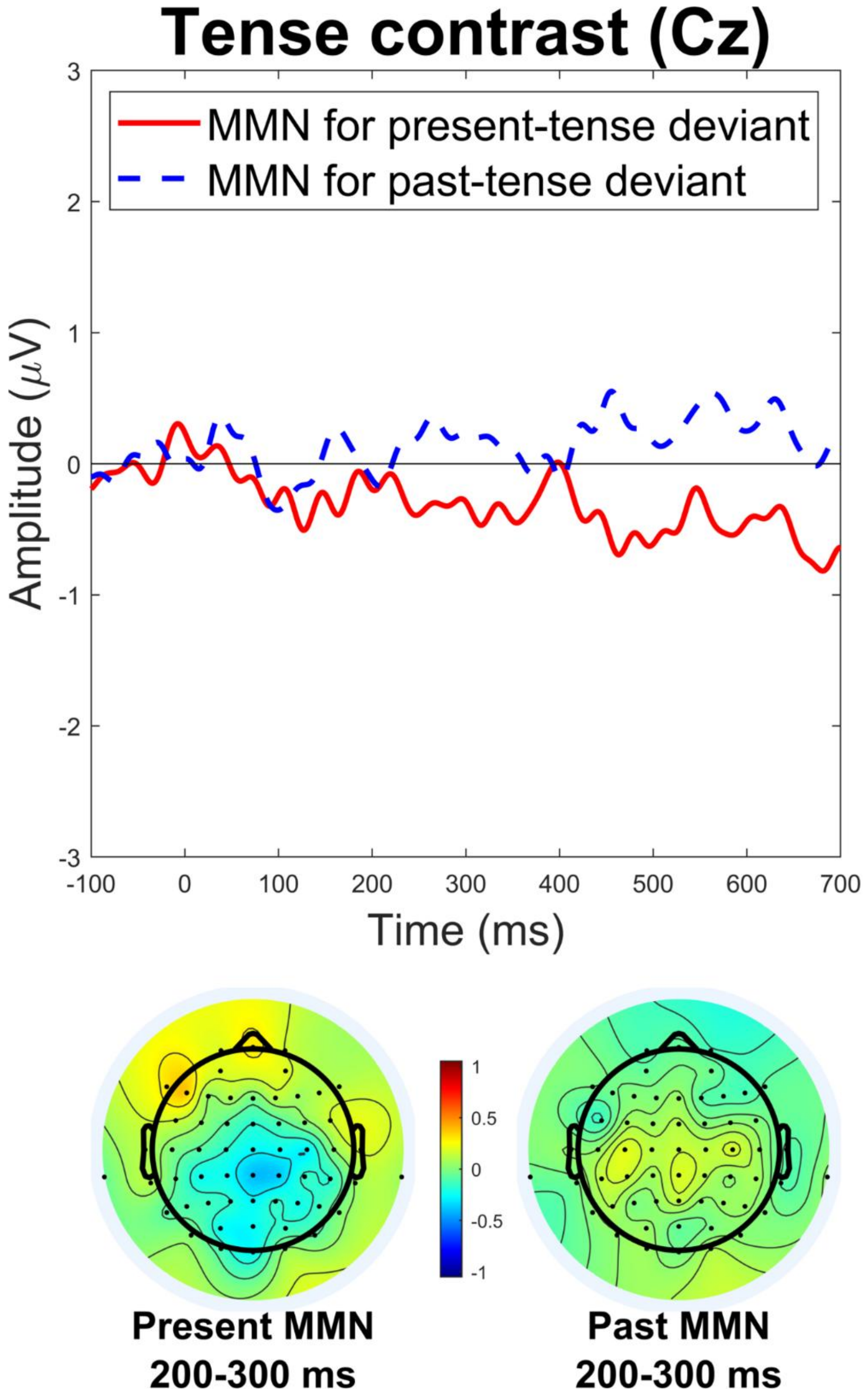
Bigger MMN

- <u>Present-tense d</u> <u>nse standards</u>
  gave ch **get** *clung pled grew met* ***pave***
  broug ng ***thank*** …

Smaller MMN

- <u>Past-tense deviants</u> <u>andards</u>
  pave choose bl cling plead grow get
  gave *b* ***ank*** …

Defaulty, underspecified?



Tense contrast (Cz)
MMN for present-tense deviant
MMN for past-tense deviant
Amplitude (μV)
3
2
1
0
-1
-2
-3
-100
0
100
200
300
400
500
600
700
Time (ms)
1
0.5
0
-0.5
-1
Present MMN
200-300 ms
Past MMN
200-300 ms

- MMN is not only sensitive to *phonological* underspecification

- It is also sensitive to underspecification (or other asymmetries in memory representation) for non-phonological contrasts

The MMN is a tool that can probe the structure of *all sorts of* linguistic representations – not just phonological ones!

gave chose bled swore ***get*** *clung pled grew met* ***thank(you)*** *brought drew sang* ***pave*** *…*